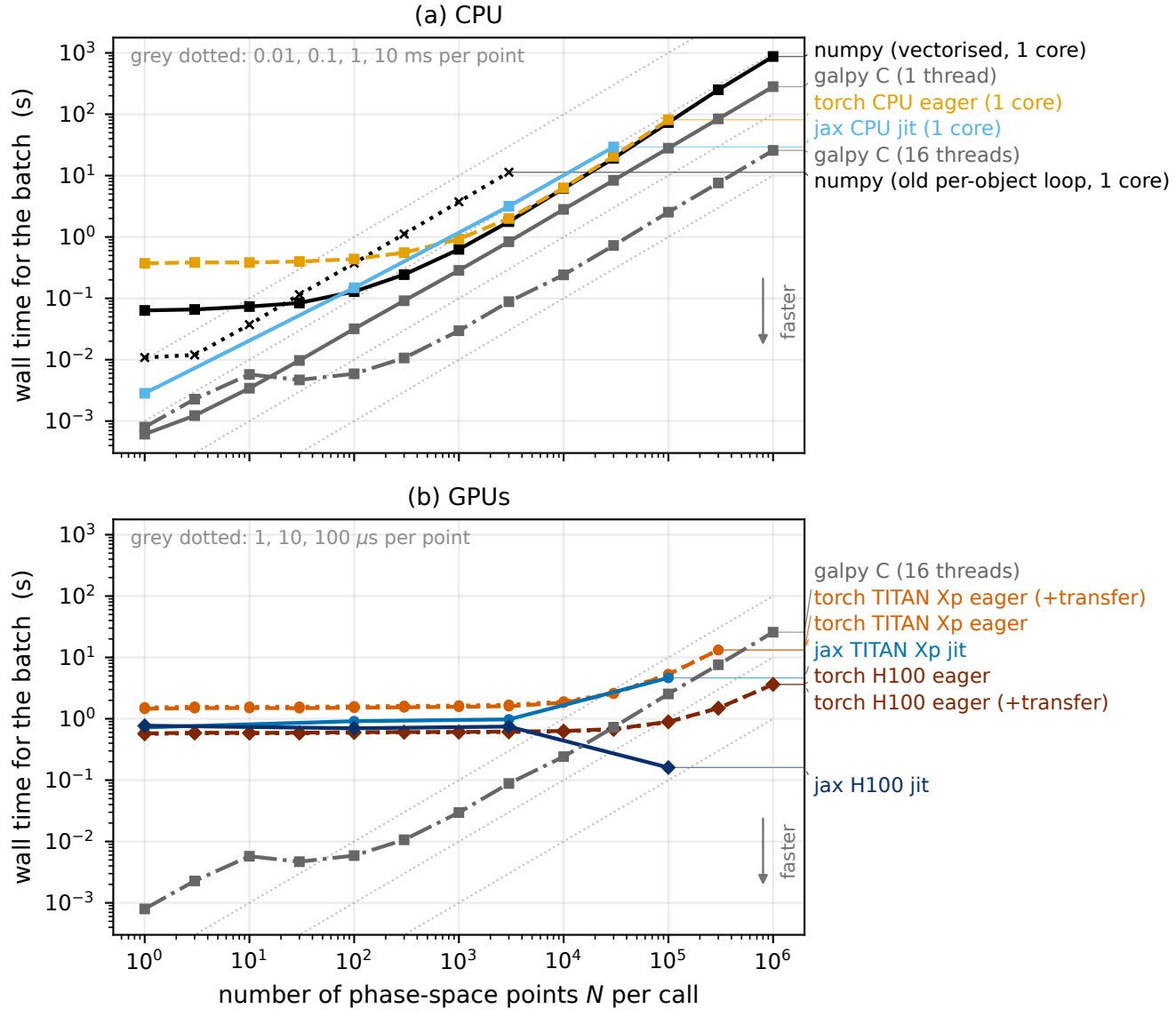
(a) CPU
grey dotted: 0.01, 0.1, 1, 10 ms per point
wall time for the batch (s)
faster
numpy (vectorised, 1 core)
galpy C (1 thread)
torch CPU eager (1 core)
jax CPU jit (1 core)
galpy C (16 threads)
numpy (old per-object loop, 1 core)
(b) GPUs
grey dotted: 1, 10, 100 μs per point
wall time for the batch (s)
number of phase-space points N per call
faster
galpy C (16 threads)
torch TITAN Xp eager (+transfer)
torch TITAN Xp eager
jax TITAN Xp jit
torch H100 eager
torch H100 eager (+transfer)
jax H100 jit